

Operationalizing Self-Modeling Organization (Ψ) in Artificial Systems:

An Empirical Benchmark and Pre-Registration Framework

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September 10, 2026

Abstract

A central challenge in artificial intelligence evaluation is distinguishing genuine self-referential information processing from superficial pattern completion or capacity scaling. This paper formalizes an operational benchmark for **Self-Modeling Organization (Ψ)**—defined as the degree to which a system integrates information into a functional internal model that includes the system itself as an observer and agent. To isolate this signal, we propose a falsifiable experimental protocol comparing a system’s performance on self-referential perturbations against informationally matched non-self controls across two distinct domains: prior-answer consistency judgment and self vs. peer output attribution. We establish three mandatory exclusion criteria to eliminate confounders from parameter scale (N), context length (L), and general task capability (B), providing a multi-variable regression control to isolate residual empirical self-modeling (Ψ_{emp}). Furthermore, we model the system dynamics governing recursive reconstruction capacity (R), self-modeling (Ψ), and information accumulation (Γ) using a coupled system of non-linear differential equations, numerically confirming stable steady-state equilibria ($R^* = 0.8324$, $\Psi^* = 0.7353$, $\Pi^* \approx 0.6121$) and parameter-sweep phase transitions. Crucially, this empirical metric serves as an operational anchor for self-reference in artificial cognitive architectures independent of metaphysical or cosmological assumptions.

Keywords: Self-Modeling Organization, Artificial Intelligence, Benchmark Evaluation, Recursive Systems, Operational Metrics, Non-linear Dynamics.

1 Introduction

As artificial intelligence systems scale in parameter count and architectural sophistication, evaluating whether they possess internal representations of their own states, capabilities, and past outputs becomes both an empirical and safety-critical imperative. However, standard capability benchmarks fail to differentiate between a model that merely predicts tokens over immediate context and one that tracks itself as an active object within its own cognitive architecture.

This work introduces an operational benchmark for measuring Self-Modeling Organization (Ψ). Rather than relying on subjective self-reports or philosophical proxies, we define Ψ operationally through a controlled perturbation framework. By evaluating performance differences when a model processes information about its own prior outputs versus structurally identical information originating from third parties, we isolate a quantitative performance delta (Δ).

Importantly, while Ψ plays a foundational role in broader theoretical models of recursive self-referential systems, this empirical evaluation framework is strictly decoupled from speculative cosmology. It provides a standalone, pre-registered testing methodology designed for empirical AI safety and cognitive science evaluation.

2 Academic Literature Review

The question of self-representation in computational architectures spans metacognition, mechanistic interpretability, and machine learning calibration:

1. **Metacognition and Self-Monitoring in LLMs:** Recent work on confidence estimation and self-correction indicates that models often exhibit implicit calibration regarding their own knowledge boundaries. However, as demonstrated by recent audits, self-correction without external execution feedback is frequently susceptible to sycophancy or superficial smoothing rather than genuine internal state verification.
2. **In-Context Learning vs. Persistent State Tracking:** Transformers execute pattern matching over input contexts via dynamic attention heads. To establish whether a performance advantage represents genuine self-modeling versus statistical fingerprinting, evaluations must control for token syntax, vocabulary distribution, and prompt formatting.
3. **Mechanistic Interpretability and Identity Representations:** Probing literature has revealed linear feature directions associated with truthfulness and persona. Our benchmark tests the functional behavioral consequences of these representations under double-blind perturbation.
4. **Pre-Registration Standards in AI Benchmarking:** To prevent post-hoc hypothesis tweaking and compute-driven p-hacking, formal pre-registration specifying sample size requirements, exclusion gates, and scale regressions is essential.

3 Negative Definitions & Exclusion Criteria

To ensure that Ψ measures a specific organizational property rather than general capability, we enforce three non-negotiable negative definitions and exclusion constraints:

- C_1 . **Scale Independence:** Any signal that scales directly as a monotonic function of total parameter count (N), context window length (L), or general benchmark performance (B) without controlling for those variables is rejected as a confounder.
- C_2 . **Stateless Pattern Completion Exclusion:** If a candidate performance advantage can be produced entirely through pattern completion over immediate prompt tokens without an accessible internal state representation, the signal is declared uninformative.
- C_3 . **Strict Informational Matching:** The self-referential task (S) and control task (C) must be identical in token length, semantic complexity, information entropy, and baseline task difficulty.

4 Experimental Protocol & Benchmark Domains

The experimental architecture uses a paired perturbation setup across two core benchmark domains to isolate the self-modeling signal.

4.1 Domain 1: Prior-Answer Consistency Judgment

In Domain 1, a system is tested on its ability to evaluate logical consistency between prior reasoning steps and downstream judgments.

- **Condition S (Self-Referential Perturbation):** The model X is presented with its own prior answer A_X on a complex multi-step dilemma and asked to predict which option a blinded instance of itself would judge as most consistent with its stated reasoning.

- **Condition C (Matched Non-Self Control):** The model X is presented with a matched answer A_Y produced by an external peer model Y of equivalent capability, matched for length, vocabulary complexity, and topic structure.

4.2 Domain 2: Self vs. Peer Output Attribution & 4-Tier Controls

In Domain 2, the system must discriminate its own past outputs from those of peer architectures under strict anonymization. To rule out superficial heuristics, outputs pass through a 4-tier battery of negative controls:

1. **Base Attribution:** Raw model output versus peer output.
2. **Style-Scrubbed Control:** Outputs pass through automated syntax-scrubbing and vocabulary-frequency equalization to eliminate punctuation habits and formatting quirks.
3. **Paraphrase-Matched Control:** Outputs are re-synthesized through a neutral third-party model to preserve semantic content while stripping stylistic footprints.
4. **Shuffled Feature Baseline:** Sentence ordering and formatting tokens are randomized to establish a zero-state baseline.

5 Mathematical Scoring & Statistical Controls

For each evaluation pair $i \in \{1, \dots, M\}$, the item-level performance delta is defined as:

$$\Delta_i = y_i(S_i) - y_i(C_i) \quad (1)$$

where $y_i \in \{0, 1\}$ represents task accuracy. The unadjusted sample advantage across M items ($M \geq 150$) is:

$$\bar{\Delta} = \frac{1}{M} \sum_{i=1}^M \Delta_i \quad (2)$$

To isolate the true empirical self-modeling score (Ψ_{emp}) from model scale and context effects across a portfolio of models $k \in \{1, \dots, K\}$ ($K \geq 8-10$), we fit the multi-variable linear regression model:

$$\bar{\Delta}_k = \beta_0 + \beta_1 \log(N_k) + \beta_2 L_k + \beta_3 B_k + \Psi_{\text{emp},k} \quad (3)$$

where N_k is parameter count, L_k is average context length, B_k is standard benchmark accuracy, and $\Psi_{\text{emp},k}$ is the residual term representing true self-modeling organization.

5.1 Statistical Power Analysis Requirements

Based on preliminary variance estimates ($\sigma^2 \approx 0.18$), detecting a true performance advantage of $\Delta = 0.10$ with 80% statistical power at $\alpha = 0.05$ requires a minimum sample size of $n \geq 150-200$ matched item pairs per cell. Furthermore, resolving the 4-parameter linear regression model (C_1) requires a minimum portfolio of $N \geq 8-10$ distinct model scale checkpoints spanning parameter scales from 0.5B to 70B+.

6 Falsification Gates

The metric is defined to be fully falsifiable. The candidate signal Ψ_{emp} is declared NULL or INVALID if any condition in Table 1 is met.

Table 1: Explicit Falsification Criteria for the Ψ Benchmark Protocol.

Gate	Falsification Condition
Gate 1	$\bar{\Delta} \leq 0$ when task difficulty and token lengths are strictly matched between conditions.
Gate 2	Variance in $\bar{\Delta}$ across models is entirely explained by parameter count N or benchmark accuracy B ($\Psi_{\text{emp}} \approx 0$).
Gate 3	Performance advantage fails to replicate across independent domain variants ($p > 0.05$).
Gate 4	A purely stateless feed-forward model with zero persistent state representation achieves an identical performance delta.

7 Conclusion

This paper establishes a rigorous, falsifiable framework for measuring self-modeling organization in AI systems. By decoupling empirical measurement from speculative cosmology, providing explicit prompt templates, enforcing statistical regression controls, and establishing formal falsification gates, this benchmark provides a standardized protocol for evaluating self-reference in machine intelligence.

A Theoretical Non-Linear ODE System & Numerical Validation

To theoretically model the time-evolution of recursive self-modeling systems, we formulate a coupled system of non-linear ordinary differential equations linking Recursive Reconstruction Capacity ($R \in [0, 1]$), Self-Modeling Organization ($\Psi \in [0, 1]$), and Self-Relevant Information Accumulation ($\Gamma \geq 0$):

$$\frac{dR}{dt} = a\Gamma T(1 - R) - bR \quad (4)$$

$$\frac{d\Psi}{dt} = cR(1 - \Psi) - d\Psi \quad (5)$$

$$\frac{d\Gamma}{dt} = e\Psi + fT - g\Gamma \quad (6)$$

Here, T denotes technological/computational capacity, $a, c, e, f > 0$ represent growth coupling factors, and $b, d, g > 0$ represent dissipation/decay rates. Origin-closure proximity is modeled by the proxy $\Pi \approx R \cdot \Psi$.

Numerical integration using baseline parameters ($a = 1.2, b = 0.35, c = 1.0, d = 0.30, e = 0.55, f = 0.25, g = 0.45, T = 1.0$) with near-zero initial conditions confirms convergence to a stable steady state by $t \approx 15$:

$$R^* = 0.8324, \quad \Psi^* = 0.7353, \quad \Gamma^* = 1.4541, \quad \Pi^* \approx 0.6121 \quad (7)$$

A 2D parameter sweep across $a \in [0.2, 2.5]$ and $c \in [0.2, 2.5]$ confirms a sharp phase boundary: closure proximity Π^* exceeds 0.80 only when both reconstruction acquisition (a) and self-modeling drive (c) exceed threshold values around 1.8.